

Activity 1: Hotel Reservation System

The Scenario / Questions: You are hired to create a database for a hotel. Follow these steps carefully:

1. **Create Database:** Create a database named HotelDB and use it.
2. **Create Guests Table (Parent 1):**
 - GuestID: Integer, must generate numbers automatically, Primary Key.
 - FullName: Varchar(100), cannot be empty.
 - PassportNo: Varchar(20), must be unique.
 - Gender: Must only accept 'Male', 'Female', or 'Other' (Use ENUM).

```
Gendar ENUM('Male', 'Female', 'Other');
```

3. **Create Rooms Table (Parent 2):**
 - RoomNo: Integer, Primary Key (No auto-increment, hotel rooms have specific numbers like 101, 102).

//auto increment - manual

```
create table Users (  
    user_id INT auto_increment primary key,  
    user_name VARCHAR(100)  
)auto_increment = 101;
```

- RoomType: Must only accept 'Standard', 'Deluxe', or 'Suite' (ENUM).
- PricePerNight: Decimal, must be greater than 0.

```
PricePerNight DECIMAL(10,2) check(pricepernight>0);
```

- Availability: Varchar(20), default value should be 'Available'.

4. **Create Bookings Table (Child / Junction Table):**

- GuestID: Integer.
- RoomNo: Integer.
- CheckInDate: Date.
- Make both GuestID and RoomNo a **Composite Primary Key**.
- Add Foreign Keys to both GuestID and RoomNo referencing their parent tables. Apply **ON DELETE CASCADE** and **ON UPDATE CASCADE** to both.

5. **Insert Data:** Insert 2 Guests, 2 Rooms, and make 2 Bookings.

Answer :

```
-- Step 1: Create and use database  
CREATE DATABASE HotelDB;  
USE HotelDB;  
  
-- Step 2: Create Guests Table (Using AUTO_INCREMENT, UNIQUE, ENUM, NOT NULL)  
CREATE TABLE Guests (  
    GuestID INT AUTO_INCREMENT PRIMARY KEY,  
    FullName VARCHAR(100) NOT NULL,  
    PassportNo VARCHAR(20) UNIQUE,  
    Gender ENUM('Male', 'Female', 'Other')
```

```

);

-- Step 3: Create Rooms Table (Using ENUM, CHECK, DEFAULT)
CREATE TABLE Rooms (
    RoomNo INT PRIMARY KEY,
    RoomType ENUM('Standard', 'Deluxe', 'Suite'),
    PricePerNight DECIMAL(10,2) CHECK (PricePerNight > 0),
    Availability VARCHAR(20) DEFAULT 'Available'
);

-- Step 4: Create Bookings Table (Using Composite Key & Cascading)
CREATE TABLE Bookings (
    GuestID INT,
    RoomNo INT,
    CheckInDate DATE,
    PRIMARY KEY (GuestID, RoomNo), -- Composite Primary Key
    FOREIGN KEY (GuestID) REFERENCES Guests(GuestID) ON DELETE CASCADE ON UPDATE
    CASCADE,
    FOREIGN KEY (RoomNo) REFERENCES Rooms(RoomNo) ON DELETE CASCADE ON UPDATE
    CASCADE
);

-- Step 5: Insert Data
-- Notice we don't insert GuestID because of AUTO_INCREMENT
INSERT INTO Guests (FullName, PassportNo, Gender) VALUES
('John Doe', 'N123456', 'Male'),
('Jane Smith', 'N987654', 'Female');

-- Notice we don't insert Availability for Room 101, it will take the DEFAULT value
INSERT INTO Rooms (RoomNo, RoomType, PricePerNight) VALUES
(101, 'Standard', 5000.00);

INSERT INTO Rooms (RoomNo, RoomType, PricePerNight, Availability) VALUES
(205, 'Suite', 15000.00, 'Booked');

-- Make bookings using the generated GuestIDs (1 and 2) and RoomNos
INSERT INTO Bookings (GuestID, RoomNo, CheckInDate) VALUES
(1, 101, '2023-12-01'),
(2, 205, '2023-12-05');

```

Activity 2: IT Company Project Management

The Scenario / Questions: Design a database to track employees and the projects they work on.

1. **Create Database:** Create CompanyDB and use it.
2. **Create Projects Table (Parent 1):**
 - ProjectCode: Integer, Primary Key.
 - ProjectName: Varchar(100), cannot be empty.
 - Budget: Decimal, must be at least 100,000.
3. **Create Employees Table (Parent 2):**
 - EmpID: Integer, Auto-increment, Primary Key.

- Email: Varchar(100), must be unique and cannot be empty.
- JobRole: Must be one of: 'Developer', 'Designer', 'Manager' (ENUM).
- Status: Must be one of: 'Active', 'OnLeave' (ENUM), Default is 'Active'.

4. Create Assignments Table (Child / Junction Table):

- EmpID: Integer.
- ProjectCode: Integer.
- AssignedHours: Integer, default is 40.
- Create a **Composite Primary Key** using EmpID and ProjectCode.
- Add a Foreign Key for EmpID with **ON DELETE CASCADE**.
- Add a Foreign Key for ProjectCode with **ON UPDATE CASCADE**.

5. Action Task: Insert some data, then write a query to **DELETE** an employee and explain what happens to their assignments.

Answer :

```
-- Step 1: Create and use database
CREATE DATABASE CompanyDB;
USE CompanyDB;

-- Step 2: Create Projects Table (Using CHECK)
CREATE TABLE Projects (
    ProjectCode INT PRIMARY KEY,
    ProjectName VARCHAR(100) NOT NULL,
    Budget DECIMAL(12,2) CHECK (Budget >= 100000.00)
);

-- Step 3: Create Employees Table (Using AUTO_INCREMENT, UNIQUE, NOT NULL, ENUM, DEFAULT)
CREATE TABLE Employees (
    EmpID INT AUTO_INCREMENT PRIMARY KEY,
    Email VARCHAR(100) UNIQUE NOT NULL,
    JobRole ENUM('Developer', 'Designer', 'Manager'),
    Status ENUM('Active', 'OnLeave') DEFAULT 'Active'
);

-- Step 4: Create Assignments Table (Using Composite Key, Default, and mixed Cascading)
CREATE TABLE Assignments (
    EmpID INT,
    ProjectCode INT,
    AssignedHours INT DEFAULT 40,
    PRIMARY KEY (EmpID, ProjectCode),
    FOREIGN KEY (EmpID) REFERENCES Employees(EmpID) ON DELETE CASCADE,
    FOREIGN KEY (ProjectCode) REFERENCES Projects(ProjectCode) ON UPDATE CASCADE
);

-- Step 5: Insert Data & Test Cascade
INSERT INTO Projects (ProjectCode, ProjectName, Budget) VALUES
(500, 'Website Redesign', 150000.00),
(501, 'Mobile App', 300000.00);

-- Inserting Employees (EmpID will be 1 and 2 automatically)
INSERT INTO Employees (Email, JobRole) VALUES
('dev1@company.com', 'Developer'),
```

```
('designer@company.com', 'Designer');

-- Assigning Employees to Projects
INSERT INTO Assignments (EmpID, ProjectCode) VALUES
(1, 500),
(1, 501),
(2, 500);

-- ACTION TASK: Deleting Employee 1
DELETE FROM Employees WHERE EmpID = 1;

-- EXPLANATION (For Students):
-- Because we used 'ON DELETE CASCADE' on the EmpID foreign key,
-- when we deleted Employee 1, all of Employee 1's records in the
-- 'Assignments' table were automatically deleted as well.
```

1. Write a query to display all Book Titles in **UPPERCASE**.
2. Write a query to display the Author's name and the **number of characters** (length) in their name.

3. We need to create a short code for books. Write a query to display the **first 3 letters** of the Book Title. (Hint: Use `LEFT()` or `SUBSTRING()`)

Activity 2: Student Exam Results (Math Functions)

Focus: `ROUND()`, `CEIL()`, `FLOOR()`

Setup Code:

```
SQL
CREATE TABLE ExamResults (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    Marks DECIMAL(5,2)
);

INSERT INTO ExamResults (StudentID, StudentName, Marks) VALUES
(101, 'Nimal', 85.75),
(102, 'Kamal', 92.40),
(103, 'Sunil', 74.99);
```

Questions for Students:

1. The school wants to round the marks to the nearest whole number. Write a query using the **ROUND()** function to display the StudentName and their rounded marks.
2. The science teacher wants to give extra marks by rounding UP everyone's marks to the next highest whole number. Write a query using the **CEIL()** function.
3. The strict math teacher wants to round DOWN everyone's marks to the lowest whole number. Write a query using the **FLOOR()** function.

Activity 3: Employee Leave Tracker (Date Functions)

Focus: `YEAR()`, `MONTH()`, `DATEDIFF()`

Setup Code:

```
SQL
CREATE TABLE EmployeeLeaves (
    LeaveID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    StartDate DATE,
    EndDate DATE
);

INSERT INTO EmployeeLeaves (LeaveID, EmpName, StartDate, EndDate) VALUES
(1, 'Amala', '2023-10-01', '2023-10-05'),
(2, 'Ruwan', '2023-12-15', '2023-12-20'),
(3, 'Kasun', '2024-01-10', '2024-01-12');
```

Questions for Students:

1. Write a query to display the EmpName and only the **Year** they started their leave.
2. Write a query to display the EmpName and only the **Month** their leave ended.
3. Calculate the total number of days each employee was on leave. Write a query using the **DATEDIFF()** function to find the difference between EndDate and StartDate.

Activity 4: Shop Sales Summary (Aggregate Functions)

Focus: SUM(), AVG(), MAX(), MIN(), COUNT()

Setup Code:

```
SQL
CREATE TABLE DailySales (
    SaleID INT PRIMARY KEY,
    ItemName VARCHAR(50),
    Amount DECIMAL(10,2)
);

INSERT INTO DailySales (SaleID, ItemName, Amount) VALUES
(1, 'Laptop', 150000.00),
(2, 'Mouse', 2500.00),
(3, 'Keyboard', 4500.00),
(4, 'Monitor', 35000.00),
(5, 'USB Drive', 1500.00);
```

Questions for Students:

1. The shop owner wants to know the **Total** sales amount for the day. Write a query using SUM().
2. Find the **Average** (AVG) price of the items sold today.
3. Write a single query to find both the **Highest** (MAX) sale amount and the **Lowest** (MIN) sale amount.
4. Count how many total items (transactions) were sold today using the COUNT() function.